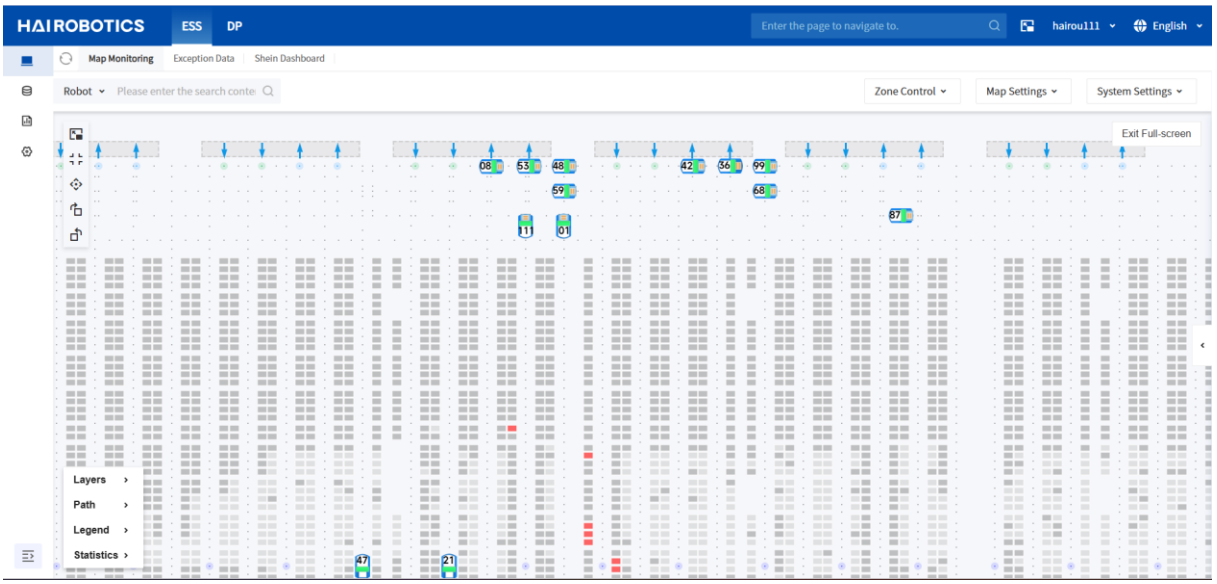
Hai Robotics ESS System Beginner's Guide

A. Understanding Icons



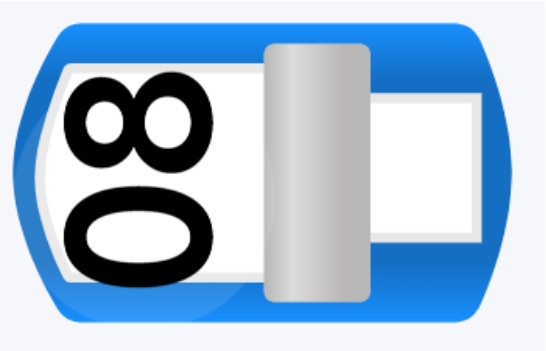
Robot Icon

This icon represents a large vehicle.

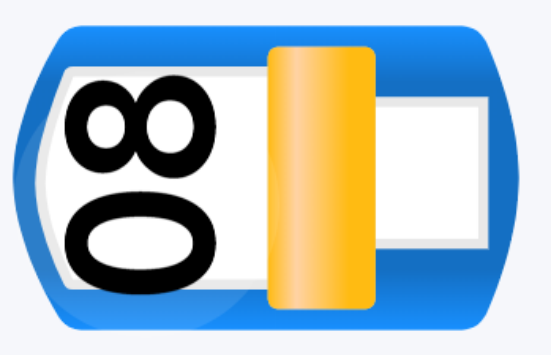
The number below represents the vehicle number, while the middle section indicates the signal light. Vehicles with fewer blank spaces represent the front end, as shown in the image below for vehicle number **03**.

When the signal light is green, it indicates that the large vehicle is operating normally on he line but has no assigned task.

When the signal light is red, it indicates a robot malfunction that requires immediate attention.



When the signal light is gray, it indicates offline control/manual control of the robot.

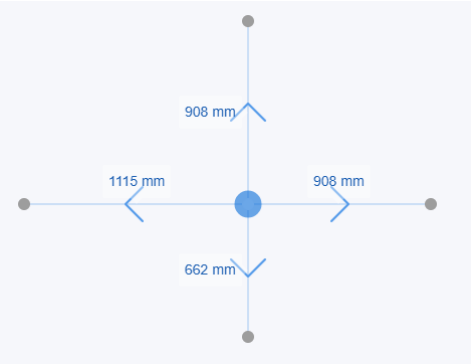


When the signal light turns yellow, it indicates that the robot will pause its operation. Manual intervention is required to trigger the pause.

If there are two yellow lines beneath the signal light, it indicates the robot has a system task assigned to it.

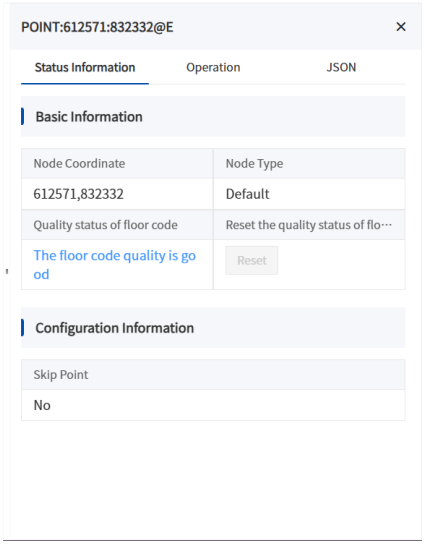
The compact vehicle's signal lights resemble those of larger vehicles, with the side featuring more blank space indicating the front end/charging port. In the diagram, the compact vehicle's front end faces left.

Ground Code Icon



Clicking the ground code icon displays the theoretical distance to nearby ground codes. If you suspect a ground code has been incorrectly placed. You can verify by checking the distance.

Clicking the right side of the ground code will display the box below.



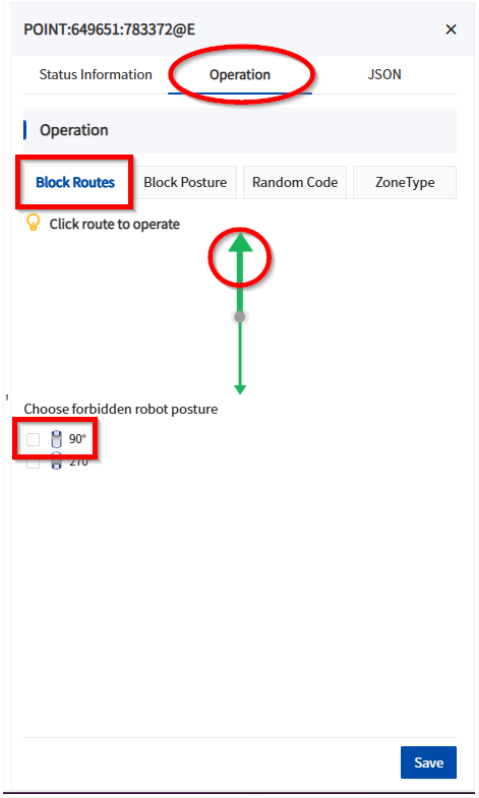
POINT:612571:832332@E: The value of the ground code. Ground codes actually applied to the ground will be rounded to this value. For the point shown above, the actual ground code values are x=**612.57** y=**832.33**.

Node Type: Rest Point (This is a rest point where robots will prioritize staying when not assigned tasks) Ground Code Quality Status: Negligible

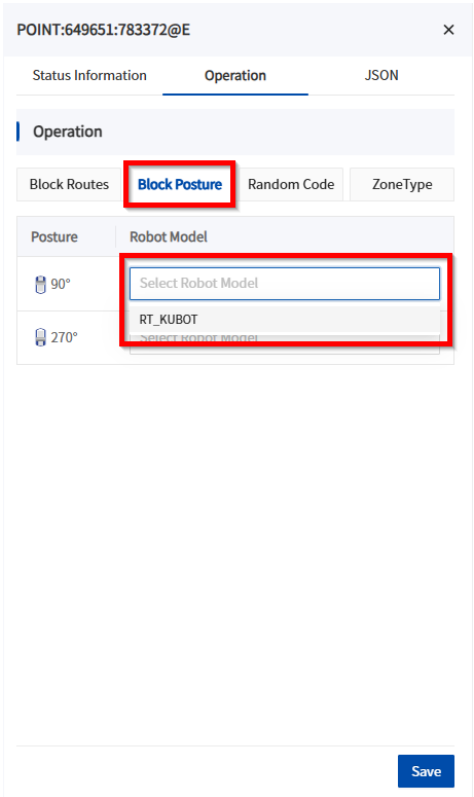
Skip Point: Yes (indicates this point is a skip point; the robot cannot pause at skip points. If certain ground codes frequently malfunction and you wish to avoid modifying the map, set this point as a skip point. The robot will then skip reading this code, preventing recognition failures due to the damaged ground code)

The point shown in the figure above serves as both a rest point and a skip point. In practice, the robot cannot rest at this point (**the skip point takes effect**). Clicking the point allows you to modify it.

1. Border Ban



Prohibited edges prevent robots from moving toward specified directions from that point. If certain routes are impassable for robots or specific points are unreachable, check the path's prohibited edges.



2. Prohibited Postures

Prohibited posture means preventing the robot from assuming a specific posture at that point. The robot has four postures: 0, 90, 180, and 270, representing its four directions.

If the robot rotates **horizontally**/**vertically** or turns sideways, it may collide with shelves or pillars. You can try prohibiting certain postures to prevent the robot from rotating/moving sideways/moving vertically at that point.

Multiple selections within the box will prohibit multiple poses (including passing through or rotating into these poses). **RT_KUBOT** Prohibits large vehicle poses

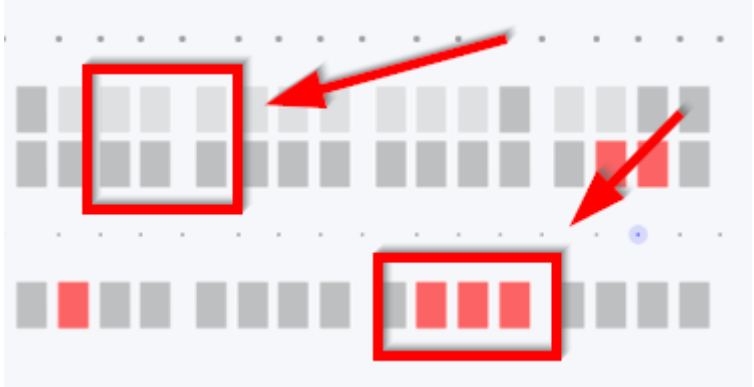
RT_KUBOT_MINI_KIVA_CT_KUBOT_EXCEPTION Prohibits carts carrying abnormal containers (those ending with “Ab”)

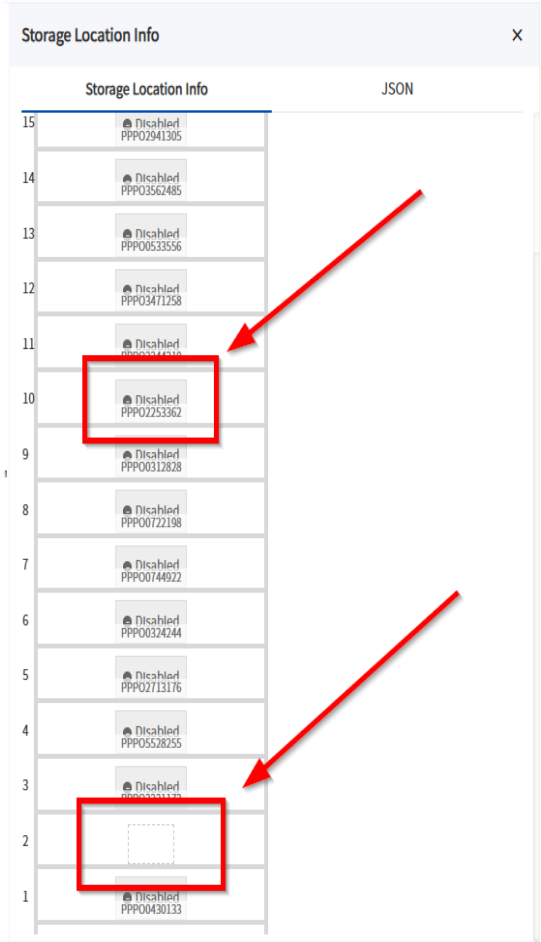
RT_KUBOT_MINI_KIVA Prohibits carts carrying empty containers

RT_KUBOT_MINI_KIVA_CT_KUBOT_STANDARD Prohibits carts carrying normal containers (loaded).

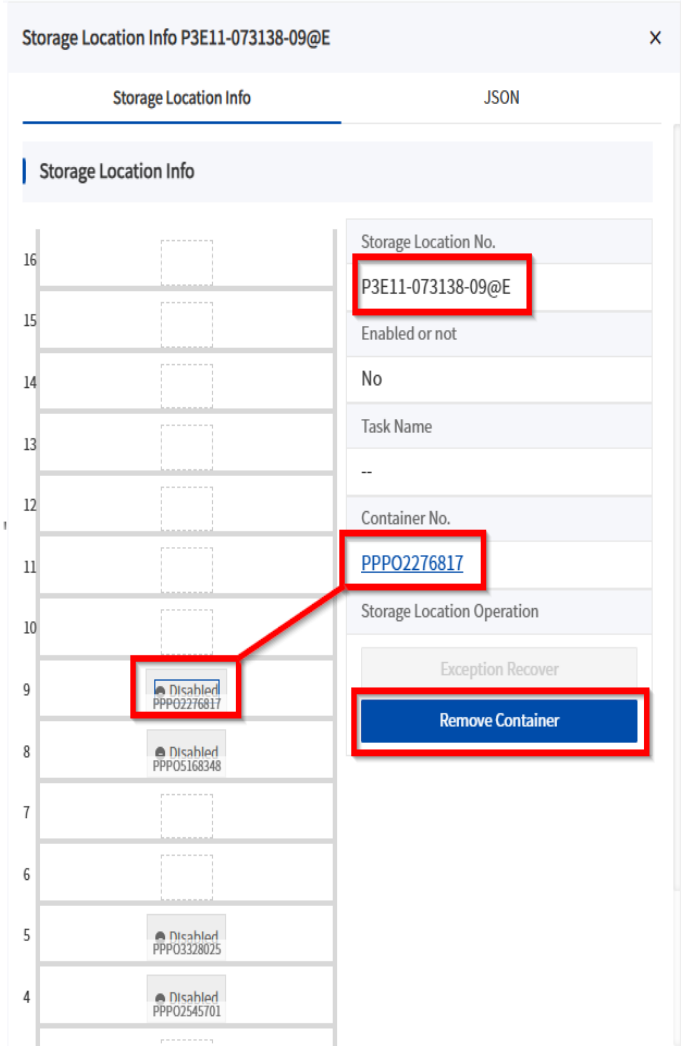
Bin Icon

The **gray** boxes on the map represent storage bays. For double-deep hook crane projects, there are deep and shallow storage bays. The **red** boxes indicate shallow storage bays (near aisles, with shallow storage positions allowing direct visibility of boxes). The **blue** boxes indicate deep storage bays (far from aisles, with deeper positions where boxes are not visible to the naked eye at higher levels).



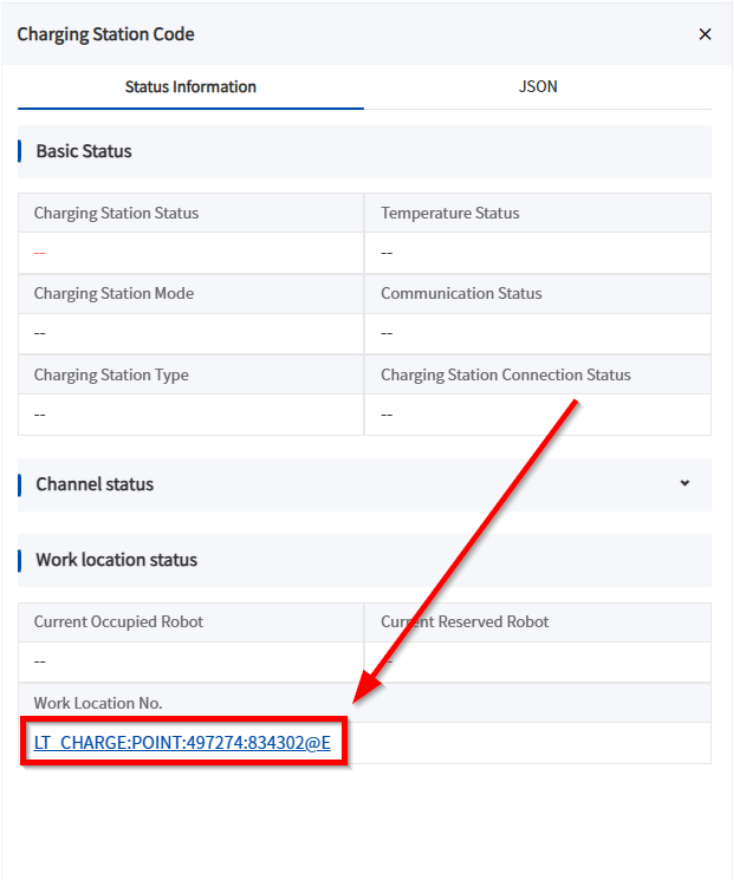


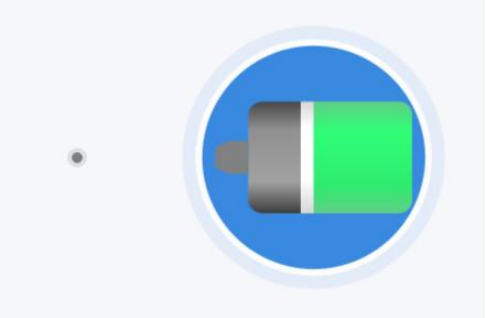
Cache slots serve as the interface between large and small vehicles. Cache slots for large vehicles (ending with “**kubot**”) and those for small vehicles (ending with “**kiva**”) reside on the same level. When one has a bin, the other automatically synchronizes.



Charging Pile Icon

Click the green charging station icon to perform operations on that charging station.





After clicking Lock, this charging station will be **disabled**, and robots will not come here to charge.

Note: Due to differences between built-in and external algorithms, or on-site configuration reasons, disabling a charging station and then unlocking it may still render it unusable. In some cases, restarting the ESS may be required for the changes to take effect.

Before locking a charging station, verify whether it can be disabled at will.

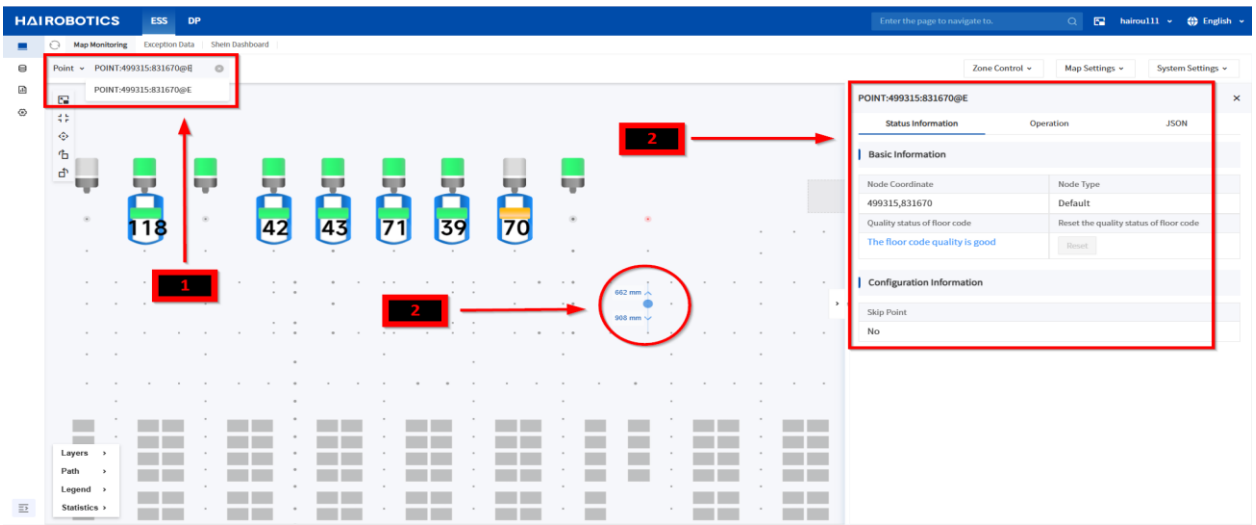
Other icons

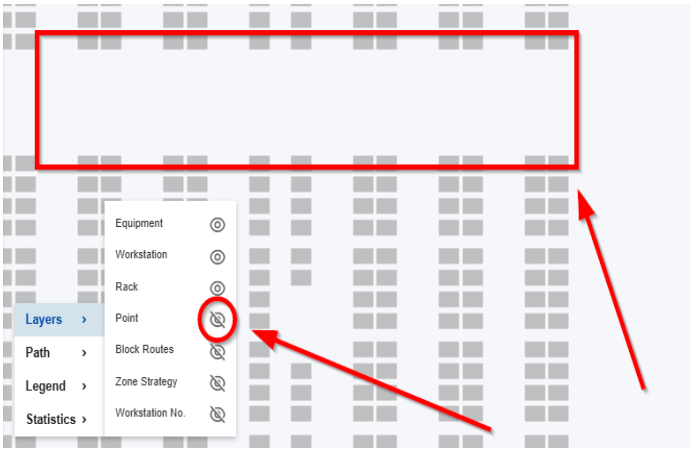
Depending on the project, additional icons such as conveyor lines, hoists, and elevators may appear. Typically, clicking these icons allows you to view the corresponding workstations. You can also access and modify them under **Workstations/Other Workstations**.

2: Button

Quick Search (Frequently Used)

In our daily work, we often need to quickly pinpoint the locations of storage bins, containers, and robots. Therefore, rapidly locating target positions is crucial.





Layers

Sometimes we want to select a specific bin location or click on a particular point. But if a large vehicle is nearby, blocking the target spot, don't just wait for the robot to move away. You can use the layer function to hide certain icons.

Common functions include:

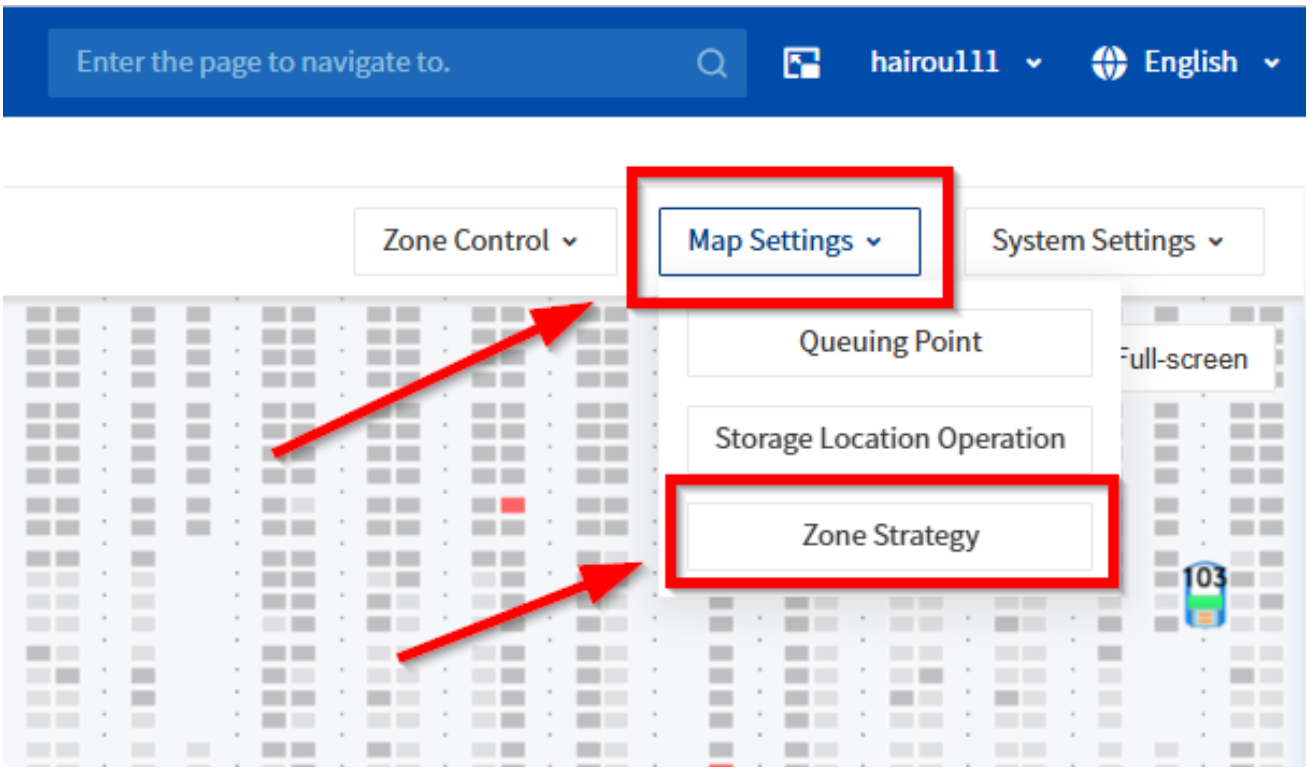
Device Masking: Enables viewing of nearby storage locations and points while preventing accidental clicks from redirecting to the robot.

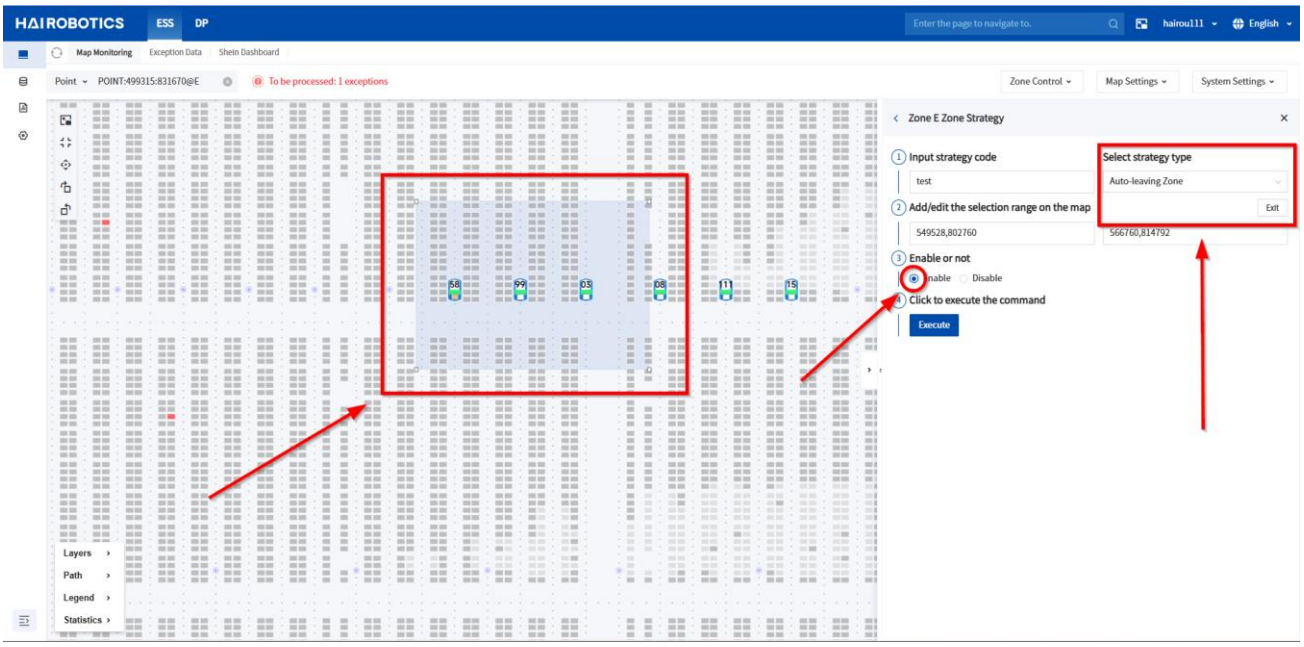
Shelf Masking: Facilitates viewing of ground code information beneath shelves.

Display Prohibited Paths: When route issues are suspected as the cause of abnormal robot picking/placing, this feature displays prohibited paths to observe inaccessible routes.

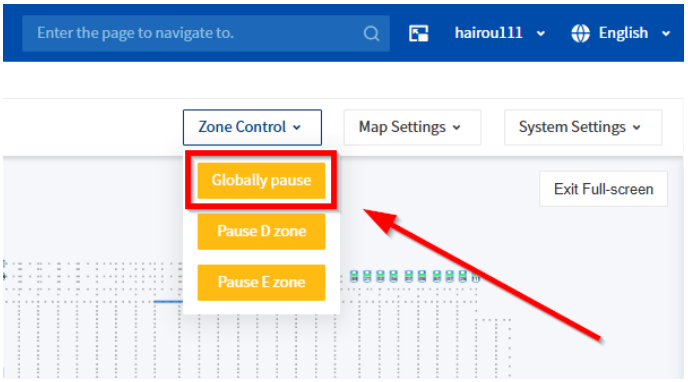
Batch Lock Storage Bins/Batch Unlock Storage Bins

In many cases, environmental factors such as shelving, floor space, and construction necessitate disabling thousands of storage locations. Manually disabling each one is clearly impractical. Below are methods for batch storage location operations.





System Pause



The pause system can halt all robotic operations. Its use should be flexible based on the scale of the project. For small projects, pauses under 30 minutes that do not impact operations may be initiated independently.

For large projects, even a 5-minute pause may disrupt operations; contact the responsible party for approval before pausing. When climbing shelves without assistance, the system must be paused.

Safe Mode

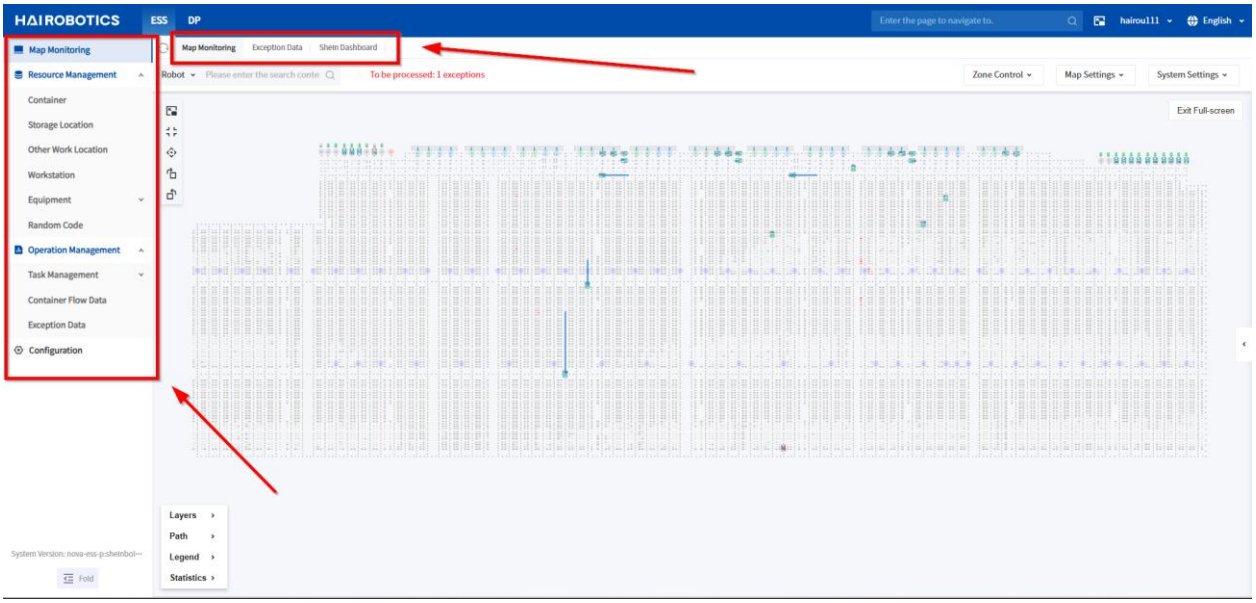
After enabling Safe Mode, all robots on the field will slow down. This has minimal impact on operations while significantly enhancing safety. We recommend using it frequently when handling exceptions during entry processing.

Other features on the map

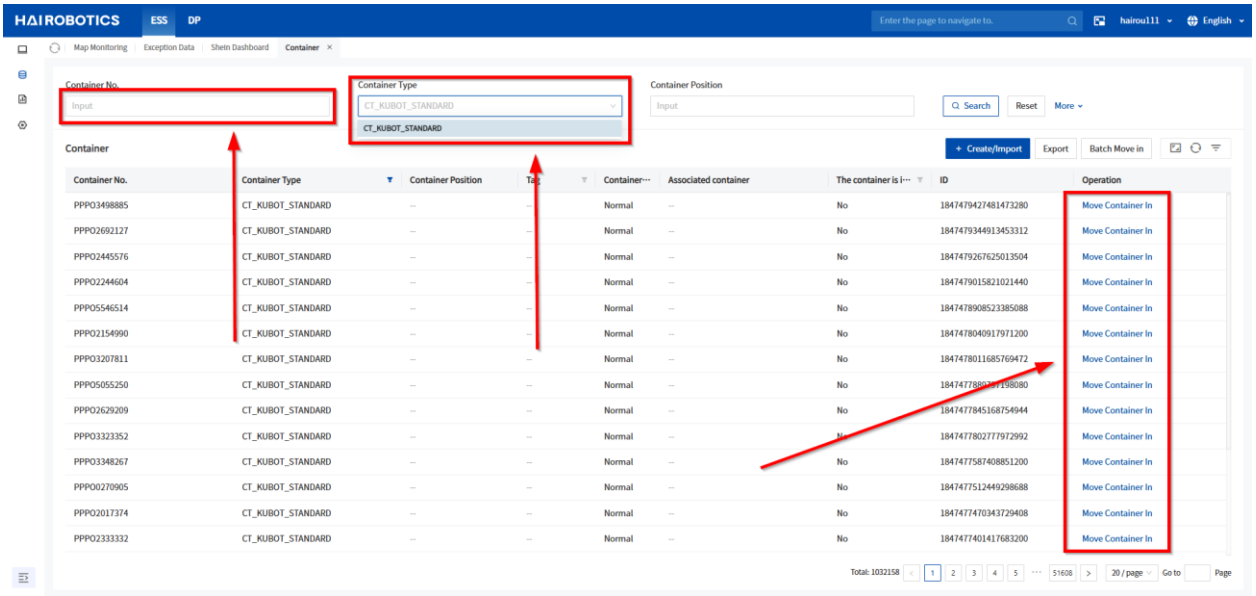
The remaining features are rarely used, so a general understanding is sufficient—there's no need to master them.

Tabs

The left side of the map features numerous tabs that can be opened simultaneously and displayed on the map's overlay layer, allowing for convenient switching between them. Many of these are essential functions that you must master.



Container (Expertise)



Create Container (Not Commonly Used)

Generally, containers are already imported when **SeaSoft** interfaces with clients, so there's no need to add non-existent containers. However, for testing purposes, a test container ending with “lee” may occasionally be created.

Container insertion/removal

To ensure the actual location of the box matches the display in the ESS system, the **move-in**/**move-out** function is frequently used. Move into storage/cache location: Simply copy the storage location number.

Loading onto the truck bed, forklift:

Fork of Cart No. 7: **kubot-7#64**

First layer of baskets on Cart No. 7: **kubot-7#0**

Second layer of baskets on Cart No. 7: **kubot-7#1**

Third layer of baskets on Cart No. 7: **kubot-7#2**

Place onto the lifting tray of the Kiva cart:

Cart No. 101: **kubot-101#64** This cart has only one location available for placing boxes. Similarly for subsequent carts.

Scenario 1:

I found a box labeled RKSS014688481 within the storage area. Upon checking the map monitoring station, I discovered the box had fallen from the **14nd** floor.

Obviously, I can't climb up to the **14nd** floor carrying the box to put it back in its original spot.

So I could try moving the container to a different storage location.

1. Search for the container on the containers page and remove it.
2. Visually locate an empty storage bin and find its bin number.
3. Move the retrieved box into the storage location you just placed it in.
4. The system shows that the box has been transferred to the designated storage location, confirming that the ESS-displayed location matches the actual physical location.

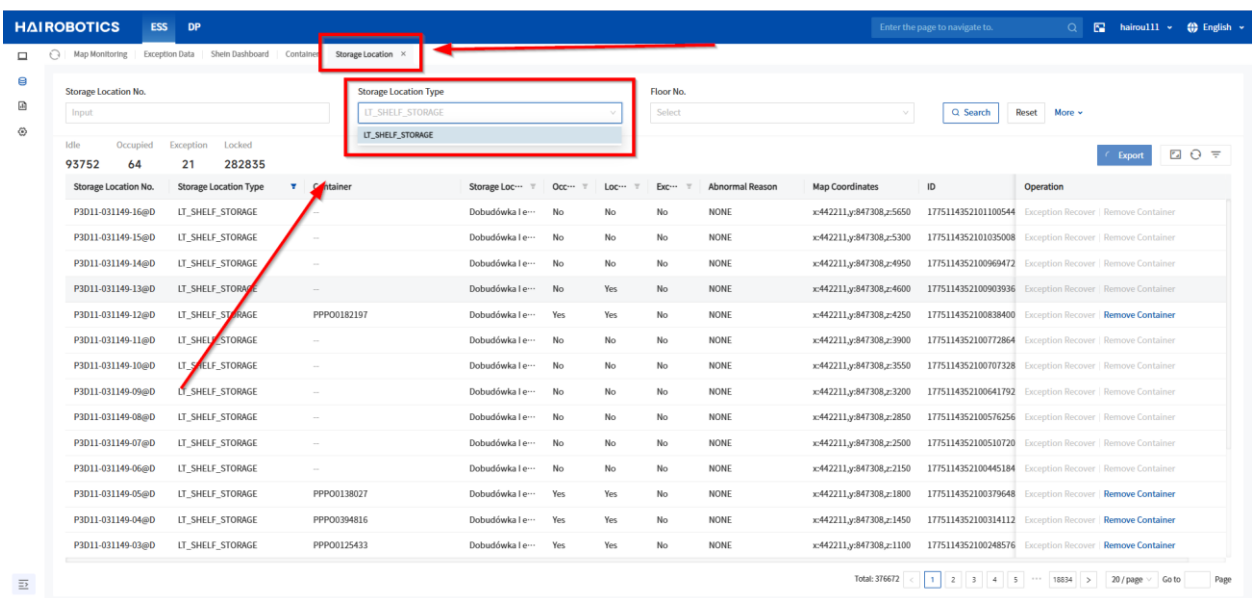
Scenario 2:

When demonstrating the maximum stacking height for large vehicles to a client, they requested that I operate the robot to place the container on the topmost level. How should I proceed?

1. Take a container and search for the box number.
2. Click “Move Container In,” enter “kubot-7#64” in the box, and the container will appear in the forklift's forks.
3. Place the actual box onto the forks, ensuring its physical position matches the location displayed in ESS.
4. Offline mode controls the crane for container placement. (For specific operations, refer to Robot—Offline Control)

Storage Location (Expertise)

Storage Location Type



- **LT_SHELF_STORAGE** represents the standard storage locations where large vehicles retrieve and place goods.

- **LT_MINI_KIVA_SHELF_STORAGE** represents the buffer location where the cart retrieves and places goods.
- **LT_KUBOT_SHELF_STORAGE** represents the buffer positions where the large vehicle retrieves and places goods.

The cache bits for both large and small cars are identical, so the quantities of the latter two must be the same!

This page requires flexible handling:

Example:

1. How many storage locations are there in total, and how many are available?
    a. Select "**LT_SHELF_STORAGE**" under "Storage Location Type." The number in the lower-right corner indicates the total storage locations.
    b. Select "**No**" under "**Occupied**" to filter out unoccupied storage locations.
2. Some boxes may be abandoned by large vehicles and marked as abnormal due to factors such as stickers on the sides or tilted boxes. If not promptly identified and addressed, this could potentially cause the box to exceed the outbound processing time limit.
    a. "Abnormal" is selected.
    b. "Occupy" is the "select" option.
    c. Abnormal storage bins with boxes can be filtered out.
    d. Screening should be conducted every 2-3 hours to ensure no abnormal boxes remain unshipped!

3. Certain cache bits were locked and disabled due to non-compliance or other reasons. Following the completion of rectification, all these cache bits must now be unlocked for normal operation.

a. Select “**LT_MINI_KIVA_SHELF_STORAGE**” or ‘**LT_KUBOT_SHELF_STORAGE**’ under “Storage Type” to filter cached storage locations.
b. Select “**Yes**” under “**Locked**” to filter locked storage locations. The resulting list shows disabled cached storage locations. Click each one individually to unlock.

Other workstations (understood)

Other workstations typically only support locking/unlocking operations, such as charging stations, drop-off points, and rest areas. You can copy coordinate points like **POINT:44474:8055** and paste them into the workstation code to filter out specific workstations.

Workstation (Learn More)

The Workstation page allows you to configure workstations and is only used when Hai Robotics provides relevant instructions. Do not modify the internal configuration under normal circumstances.

Equipment—Robots (Learn More)

1. You can determine the number of damaged robots by filtering for “Offline” status in the “Work Status” section.
2. Occasionally, you can sort by battery level to monitor the overall battery status of the robots.
3. Modify the number of backpacks configured for the robot (requires explicit instructions to proceed)

Business Tasks (Proficiency)

Business tasks are inbound and outbound inventory tasks assigned by clients, such as Shein. Our primary responsibility is to ensure these business tasks are completed smoothly.

Business Task Code: A unique identifier for a task that corresponds to only one container. When a client provides a business task,

1. When searching for a task by its task number, you can use this filter to quickly locate the task and retrieve relevant information.
2. Task Status: Typically filters only pending and in progress.

Task Type: Typically filters only for outbound handling (outbound tasks) and inbound shelving (inbound tasks).

1. The **putaway** process is completed when the cart places the bin into the buffer location. Simultaneously, a virtual storage location is assigned to the bin. At this point, the bin can be removed and transferred to any storage location, as the virtual location remains unchanged.
2. The outbound process completes once the box code is scanned on the outbound line. However, missing codes on the sides frequently cause tasks to remain in progress indefinitely. If workers can load pallets, it indicates that the task has been completed on the XY side, and tasks still in progress can be ignored.
3. The meaning of the results obtained through simultaneous screening:
    a. Pending + Transport: The task has reached the ESS system, but no large vehicle has arrived to pick it up.
    b. In progress + handling: The robot has accepted the task. If a large vehicle is handling it, the box will typically be retrieved within three to five minutes. If a Kiva cart is handling it, this indicates the cart is either retrieving the box from a buffer location or end route to the conveyor line to place the box. Delivery to the conveyor line for outbound processing usually takes 2-3 minutes. If the box has no assigned location, it may be due to an unsuccessful barcode scan. Generally, no action is required.
    c. Pending + Receiving & Shelving: The receiving task has been generated but not yet assigned to the receiving line. For XY, tasks are typically generated after “temporary storage,” with each assigned box

triggering its corresponding receiving and shelving task to change to “in progress.”

d. In Progress + Shelving: The robot is currently performing the shelving action for this box. For the XYX cart project, once the cart retrieves the box, the task status changes to “In Progress.” After placing the box in the buffer location, the task status changes to “Completed.” Business.

Business Group ID: Tasks dispatched in batches share the same Business Group ID and are generally not used as a filtering criterion.

Specified Container: Customers sometimes provide container numbers for outbound/inbound shipments. Filtering by these numbers allows for quick retrieval of related business tasks.

Creation Time: The time when the task was created

Last Updated: The time of the most recent task change (This time is updated each time the following actions occur: large vehicle task pickup, large vehicle container pickup, large vehicle container drop-off, small vehicle 24-container pickup, small vehicle container drop-off).

Source location: Ignore

Destination Position: A destination position is typically assigned only during outbound processing. It denotes the conveyor line number. For example, 3-5 indicates that the destination position for this box is the 5th section of conveyor line 3. It is essential to clearly understand the numbering of each conveyor line on-site.

Priority: The higher the priority number, the higher the priority the cart will have in executing the task.

Adjusting Priority: Typically, raising the priority of certain boxes increases their outbound speed (this operation can be used freely without causing significant issues).

1. I found a dropped box in the warehouse. How do I determine its purpose? First, look up the box number in the business tasks to see if there's a related task. If it's an active transport task, it means the box is for outbound shipment. Then check the box's current location. If it's in a buffer position,

return it to the buffer. If it's on a robot, return it to the robot. If the container has no assigned position, it indicates the cart has completed the placement action on the conveyor line (though it actually placed air onto the line since the box fell en route). You'll need to check the box's target location in the business task and manually place it onto the corresponding conveyor line.

   a. For completed putaway tasks, simply look up the bin location information and return the box to its corresponding spot. For pending putaway tasks (extremely rare), this indicates a misplaced box that wasn't scanned. You may choose to re-enter it from the receiving line.
   b. If no related task exists: This may occur if the putaway was performed too early, causing relevant records to be overwritten. Locate the bin's storage position and return it to its original spot. If no storage position information is available, place the physical bin in any available position and synchronize the container's movement to that location in ESS.

2. The client asked how many items were shipped between 8:00 and 9:00 AM and how to count them. In the task list, filter by “Task Status (Completed) + Task Type (Moving) + Update Time (8:00-9:00 AM)”. The specific number will be displayed in the bottom-right corner.

System Tasks (Mastery)

- System tasks are assignments dispatched to robots by the ESS system. These tasks may be canceled at any time (except when the cart is being stored), and the system will reassign tasks to the robot.
- If completing business tasks is the ultimate goal, then system tasks are the stepping stones leading to that goal.

To accomplish this operational task, the ESS system issues system commands to the robot, such as moving, picking up boxes, and placing boxes.

- Call-Workstation Robot Task (Hai Port Robot Box Retrieval)
- Return: Task - Return Crates to Warehouse (Hai Port crate pickup completed, return crate to warehouse task)

- **Container**:task-Bin Retrieval Task (De-shelving Task, Shelf Retrieval Task)
- **mc**:task-Bin Transfer to Workstation Task (Task to proceed to buffer shelf **waiting** point after retrieving bin from shelf)
- **charge**:task-Charging Task
- **rest**:task-Rest Task
- **pp**:task-Avoidance Task
- **Load**-Workstation Bin Retrieval Task
- **Preload-Pre-loading** Task
- **Scan-Barcode** Scanning Task

Filter system tasks with a status of "Pending," and you can resume pending tasks here. System tasks can be double-clicked to view detailed information.

The information that can be obtained from it includes:

- Storage location information for container pickup and drop-off by large/small vehicles
- By filtering through container IDs, you can determine
- Which large/small vehicle handled this container and what changes it underwent within a few days.
- The system task proves highly effective for locating lost containers.

Callback Task (Learn)

Callback tasks refer to ESS uploading completed system tasks/business tasks to clients (such as XY) via brief fields.

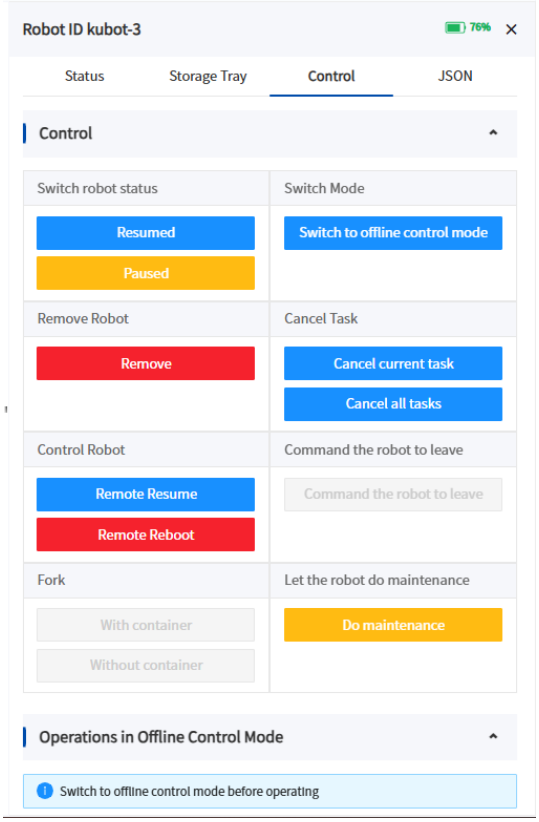
Callback tasks are critical for integrating the Hai Robotics system with client systems. Only when the client actually receives the callback notification is the outbound/inbound task considered complete in the client system. If synchronization fails between both parties, it likely indicates an issue with the task callback.

1. The client reports that a particular inbound box has not yet been shelved, while our business task records show the inbound task as completed.

2. This may indicate a failed callback transmission for the inbound task, causing asynchrony between systems. Causes vary; promptly report such incidents to Hai Robotics technical support for analysis.

If capable and with corresponding templates, retry transmissions via adapter or Postman. Alternatively, filter by container ID within the callback task and resend all callbacks sequentially (though this is often ineffective).

Robot Control



Click the robot icon, and the following page will appear on the right.

Resume/Pause: Used in combination. Pause halts the robot's actions (the robot will enter a stopped state only after completing its current system task), while Resume cancels the robot's paused state.

Switch to offline control: Manually control the robot. Offline control can only be activated when the robot has no active tasks.

Remove Robot: Removing a robot from the map will cause its icon, tasks, and any containers it carries to disappear. ESS will no longer know the robot's location. Robots are typically only removed when they become unusable due to severe malfunction. Since ESS will be unable to determine the robot's position, other robots' task planning may inadvertently route through the removed robot. Therefore, it is best to push the robot off the field before removal. The location of any boxes carried by the robot will be lost, and their status will become abnormal. Ensure proper handling of the boxes.

Cancel current task: Only cancels the system task at the front of the sequence.

Cancel All Tasks: Typically, canceling all tasks is only necessary when you wish to control the robot offline. Users often encounter failures when canceling the current task, or consistently find one task fails to cancel during a full cancellation. This occurs because system tasks currently being executed by the robot cannot be canceled! You can first pause the robot. This causes the robot to complete its current task before pausing. Subsequent tasks will remain pending and not yet executed, allowing them to be canceled entirely. Fork with/without container: If the robot restarts during a box placement/retrieval action, these buttons will illuminate after restart. Select “Fork with container” or “Fork without container” based on the actual situation.

Offline Control Instructions

- Click “Switch to Offline Control” to enter manual operation mode for the robot.
- In offline control mode, tasks are manually assigned and will not accept operational tasks from the scheduling system. (If the robot's forks, camera, or other material handling components are damaged, or if there is no space to park the vehicle, switch it to offline mode. The robot will autonomously charge and avoid other robots, but will not accept tasks.)
- Offline mode is map-dependent, preventing robots from executing impossible actions. For example, while the host computer might command a robot to

traverse aisles sideways, this would collide with racks. However, in offline mode, ground codes in aisles prohibit lateral movement. Thus, even if you issue a lateral command offline, the robot won't execute it.

- Similarly, if the current location has height restrictions and you issue a forklift lift/lower command, constrained by the dispatch system, the robot cannot exceed the height limit. Instead, it will move to an area without height restrictions to perform the lift/lower operation—a relatively intelligent approach. Therefore, feel free to experiment with offline control commands to gain familiarity. Even if you issue an unreasonable command, the robot will simply stop and remain stationary. You can then cancel the task afterward.

Move: Issue a movement command to direct the robot to a specified location and halt at the target spot in a designated posture. You may click the ground code at the destination. (If the ground code is obscured, you may first copy the point coordinates and paste them into the input field. Alternatively, you may hide the robot and shelf icons. See the “Button - Layers” section for details.)

Picking: Enter the bin number or container number to retrieve the container onto the forks. If an incorrect bin number or container number is entered, the task will fail immediately. When picking from deep bins, the robot will first retrieve containers from shallow bins, place them in the carrier, and then proceed to deep bins. Therefore, after completing the action, it is best to combine it with a place command to return the shallow bin containers.

Simply enter the box number already on the robot and the target storage location to place the box.

Fork Lift Height: Controls the height at which the main carriage forks/trolley lifts pallets.

- Main Carriage Fork Zero Height: **400mm**
- Trolley Lift Zero Height: **310mm**

Internal box retrieval: Remove the box from the basket onto the forks.

Internal box placement: Place the box from the forks into the basket.

- **Bin Detection (Understanding)**: Select a region, and the overhead crane will perform pick-and-place detection for each bin.
- **Ground Marker Inspection (Understanding)**: Select an area, and the robot will stop at each ground marker in sequence to detect any damage or missing markers.

Scenario: The customer asked me to retrieve three specific boxes, but they're stacked too high to lift out. What should I do?

1. Locate the largest vehicle with the fewest boxes on its body. When its status shows it is placing a box, click Pause. Cancel all its tasks. (Because pausing the robot causes it to complete its current action before stopping. Therefore, after pausing, wait 10-30 seconds. The box in the robot's forks will disappear, and all tasks can be canceled.)
2. Click Control, switch to Offline Control, click Pick Box, and enter the corresponding box number. Click Resume Action. (Since the robot pauses after each action, you must resume it before each task to continue.)
3. After the truck has finished picking up the containers, click “Internal Container Placement” to transfer the container from the forklift to the cargo basket.
4. Repeat steps 2-3 to remove all designated boxes in sequence.
5. Click “Move” to position the truck at the spot closest to the exit, making it easier to unload these boxes manually.
6. Enter the Kubot robot number, such as kubot-13, in the “Container Location” field within the “Container” section to filter all boxes on that large vehicle. Then remove these boxes.
7. Exit offline control and resume crane operation.